



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Class Activity – 20/07/2026

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1.Hospital Patient Admission Analysis R

Code:

```
# Install Packages (Run only once)
```

```
install.packages("ggplot2")
```

```
install.packages("treemap")
```

```
# Load Libraries
```

```
library(ggplot2)
```

```
library(treemap) #
```

```
Create Dataset hospital
```

```
<- data.frame(
```

```
Department = c(
```

```
rep("Cardiology",5),
```

```
rep("Neurology",5),
```

```
rep("Pediatrics",5),
```

```
rep("Orthopedics",5),
```

```
rep("Emergency",5)
```

```
),
```

```
Admissions = c(
```

```
120,135,140,128,150,
```

```
90,95,105,100,110,
```

```
180,170,190,200,210,
```

```

130,125,145,138,150,
240,250,260,255,270
)
)
#-----
# 1. Bar Chart
#-----
bar_data <- aggregate(Admissions~Department,hospital,sum)
ggplot(bar_data,
aes(x=Department,y=Admissions,fill=Department))+
geom_bar(stat="identity")+ labs(title="Hospital Patient
Admissions", x="Department", y="Total
Admissions")+ theme_minimal()
#-----
# 2. Stacked Bar Chart
#-----
hospital$Quarter <- rep(c("Q1","Q2","Q3","Q4","Q1"),5)
ggplot(hospital,
aes(x=Quarter,y=Admissions,fill=Department))+
geom_bar(stat="identity")+ labs(title="Department-
wise Admissions by Quarter", x="Quarter",
y="Admissions")+ theme_minimal()
#-----
# 3. Treemap
#-----
treemap(bar_data,
index="Department",
vSize="Admissions", title="Hospital

```

```
Admission Treemap") #-----
```

```
-----
```

```
# 4. Box Plot
```

```
#-----
```

```
ggplot(hospital,  
aes(x=Department,  
y=Admissions,  
fill=Department))+  
geom_boxplot()+  
labs(title="Admission Variability",  
x="Department",  
y="Admissions")+  
theme_minimal()
```

```
#-----
```

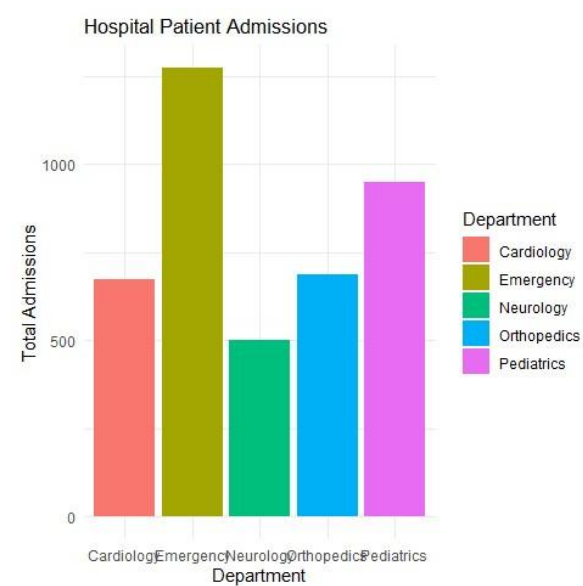
```
# 5. Histogram
```

```
#-----
```

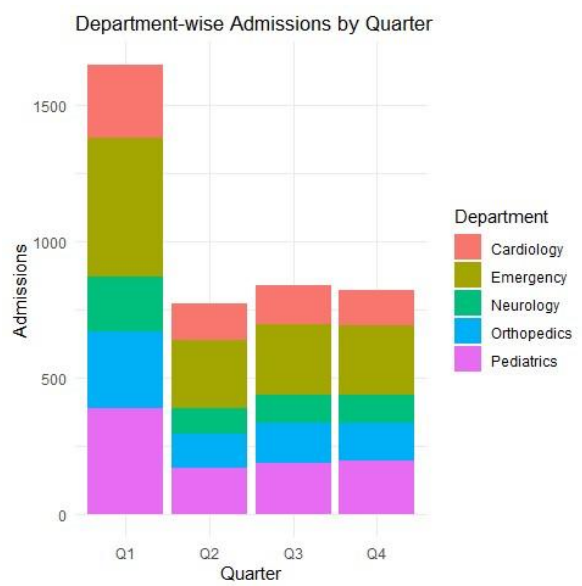
```
ggplot(hospital,  
aes(x=Admissions,  
fill=Department))+  
geom_histogram(binwidth=15,  
color="black",  
alpha=0.7)+ labs(title="Distribution of  
Admissions", x="Admissions",  
y="Frequency")+ theme_minimal()
```

Output:

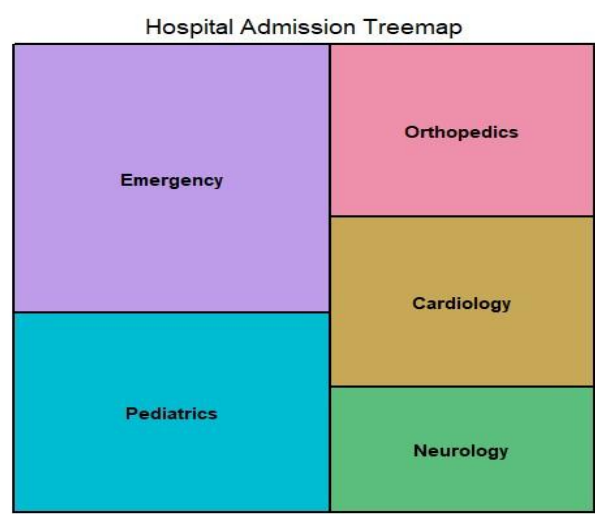
Bar Chart:



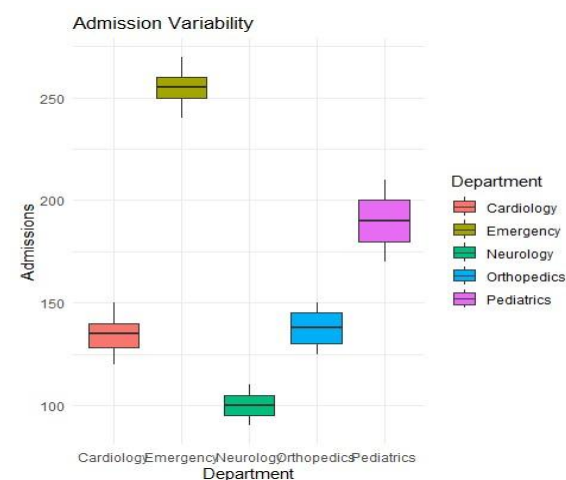
Stacked Bar Chart:



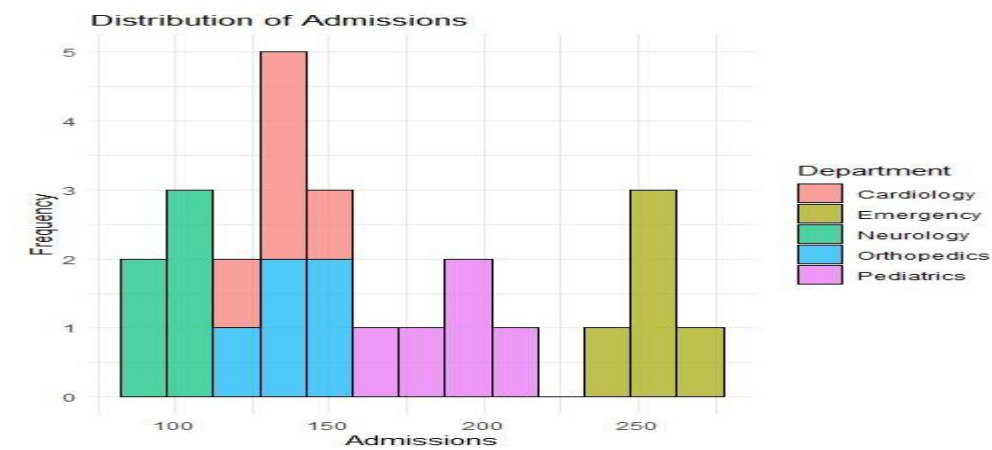
Treemap:



Box Plot:



Histogram:



1. E-Commerce Customer Spending Behaviour.

Histogram:

```
ggplot(ecommerce,  
aes(x = PurchaseAmount)) +  
geom_histogram(binwidth = 50,  
fill = "skyblue", color =  
"black") + labs(title = "Purchase Amount  
Distribution", x = "Purchase Amount",  
y = "Frequency") + theme_minimal()
```

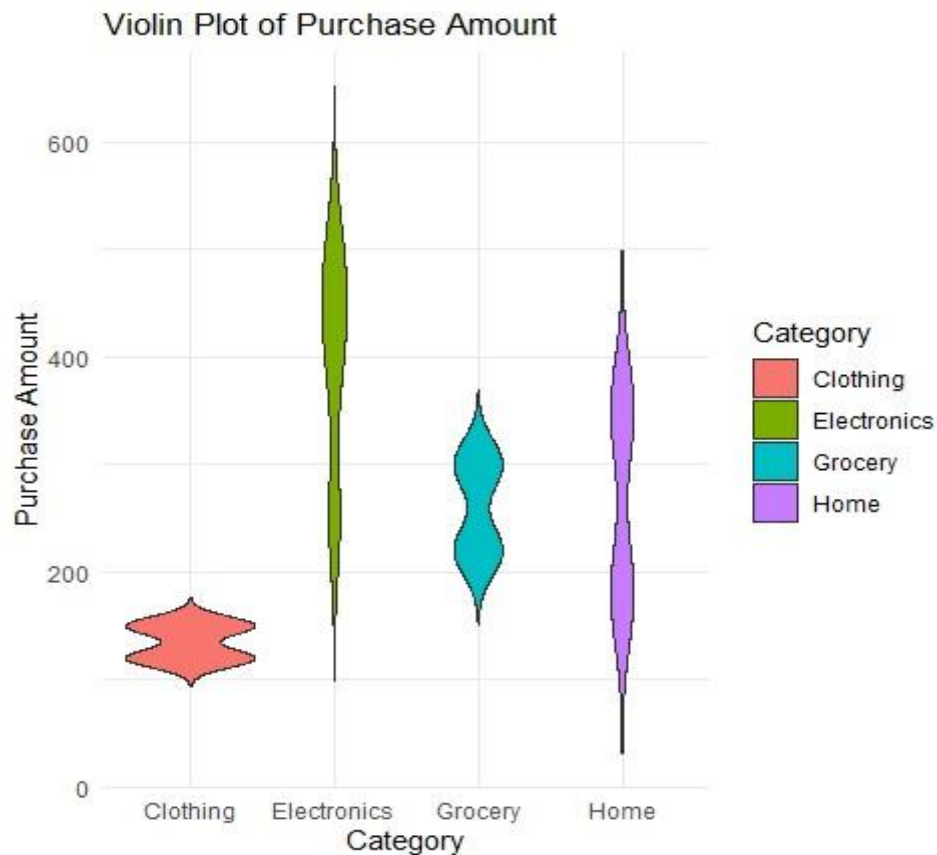
Output:



Violin Plot:

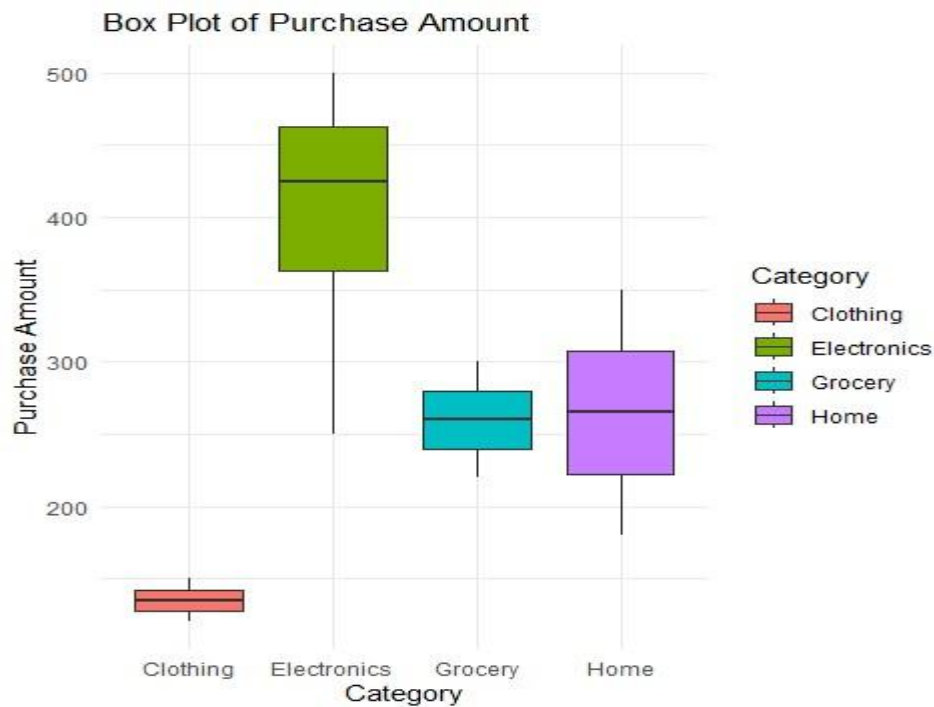
```
ggplot(ecommerce,  
aes(x = Category,  
y = PurchaseAmount,  
fill = Category)) +
```

```
geom_violin(trim = FALSE)
+ labs(title = "Violin Plot
of Purchase Amount", x
= "Category", y =
"Purchase Amount") +
theme_minimal() Output:
```

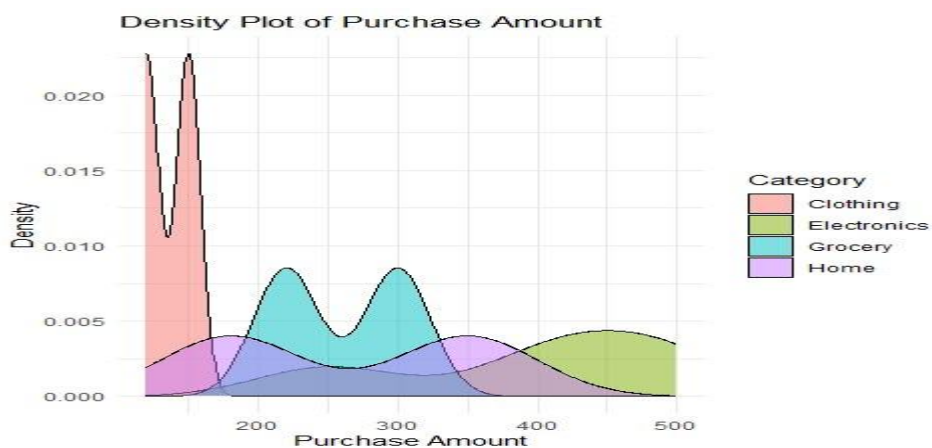


```
Box Plot ggplot(ecommerce,
aes(x = Category, y =
PurchaseAmount, fill = Category)) +
geom_boxplot() + labs(title = "Box Plot of
Purchase Amount", x = "Category",
y = "Purchase Amount") +
theme_minimal()
```

Output:



Density Plot: `ggplot(ecommerce, aes(x = PurchaseAmount, fill = Category)) +
geom_density(alpha = 0.5) + labs(title = "Density Plot of Purchase Amount", x = "Purchase Amount", y = "Density") +
theme_minimal() Output:`



Bar chart:

`ggplot(ecommerce, aes(x = Customer, y = PurchaseAmount, fill = Category)) + geom_bar(stat = "identity")`

```
+ labs(title = "Customer Purchase
Amount",    x = "Customer",    y =
"Purchase Amount") + theme_minimal()
```

Output:



2. University Examination Performance Dashboard Bar

Chart:

Install package (Run only once)

```
install.packages("ggplot2")
```

Load library

```
library(ggplot2) #
```

Create Dataset

```
students <-
```

```
data.frame(
```

```
  Student = paste0("S", 1:15),
```

```
  Department = c("CSE", "AI", "IT", "ECE", "CSE",
```

```
                "AI", "IT", "ECE", "CSE", "AI",
```

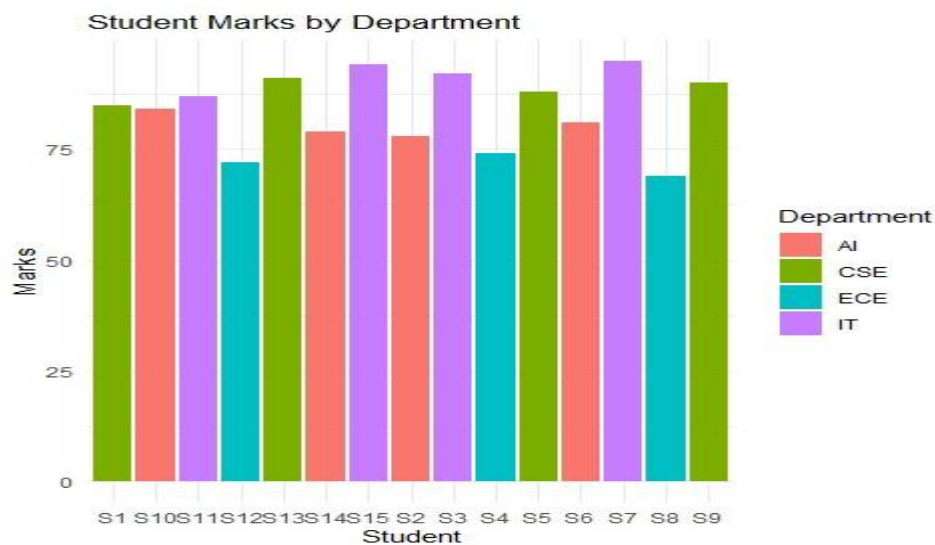
```
                "IT", "ECE", "CSE", "AI", "IT"),
```

```
  Marks = c(85, 78, 92, 74, 88,
```

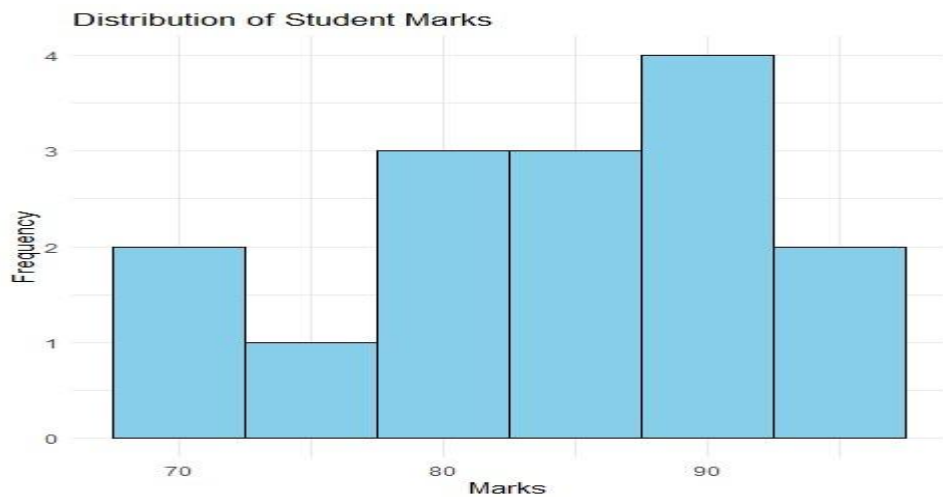
```
            81, 95, 69, 90, 84,
```

```
            87, 72, 91, 79, 94)
```

```
)
ggplot(students, aes(x = Student,
y = Marks, fill = Department)) +
geom_bar(stat = "identity") + labs(title =
"Student Marks by Department", x =
"Student", y = "Marks") +
theme_minimal() Output:
```

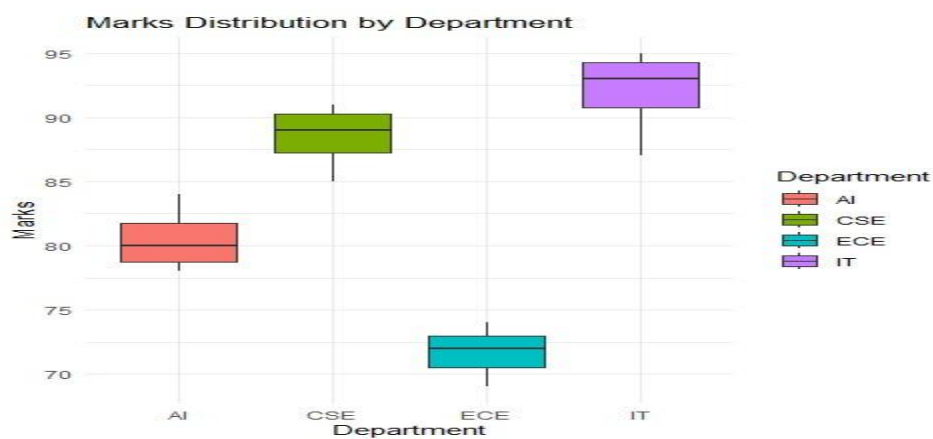


```
Histogram: ggplot(students,
aes(x = Marks)) +
geom_histogram(binwidth = 5,
fill = "skyblue", color =
"black") + labs(title = "Distribution of
Student Marks", x = "Marks", y =
"Frequency") +
theme_minimal() Output:
```

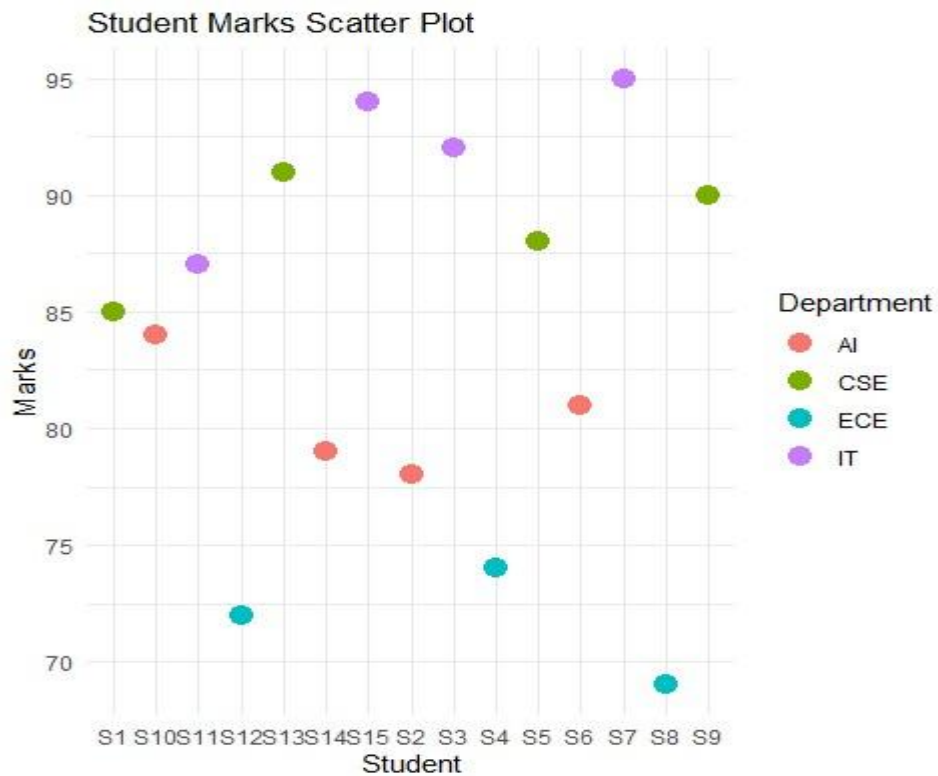


Box Plot:

```
ggplot(students, aes(x = Department,
y = Marks, fill = Department)) +
geom_boxplot() + labs(title = "Marks
Distribution by Department", x =
"Department", y = "Marks") +
theme_minimal() Output:
```



```
Scatter Plot: ggplot(students, aes(x
= Student, y = Marks, color
= Department)) + geom_point(size = 4)
+ labs(title = "Student Marks Scatter
Plot", x = "Student", y =
"Marks") + theme_minimal() Output:
```

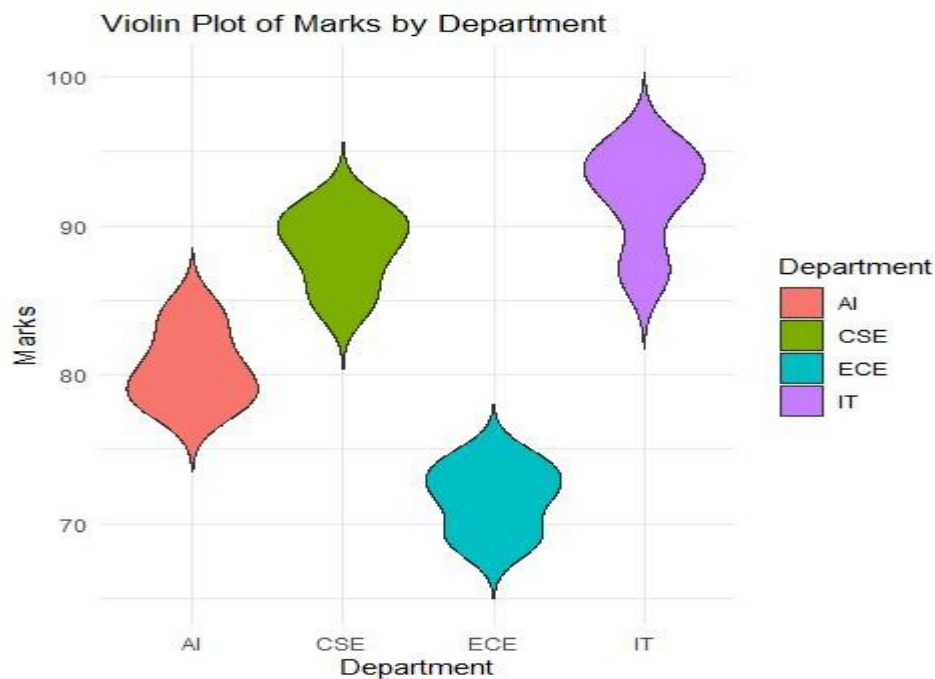


```

Violin Plot: ggplot(students,
aes(x = Department,      y =
Marks,      fill =
Department)) +
geom_violin(trim = FALSE) +
labs(title = "Violin Plot of
Marks by Department",    x
= "Department",    y =
"Marks") + theme_minimal()

```

Output:



3. Climate and Rainfall Distribution Study

Histogram:

```
# Install package (Run only once) install.packages("ggplot2")

# Load library

library(ggplot2) #

Create Dataset

climate <- data.frame(

  Month = factor(c("Jan","Feb","Mar","Apr","May","Jun",
"Jul","Aug","Sep","Oct","Nov","Dec"),
levels=c("Jan","Feb","Mar","Apr","May","Jun",
"Jul","Aug","Sep","Oct","Nov","Dec")),

  Rainfall = c(25,30,40,55,120,210,320,290,180,100,60,35),

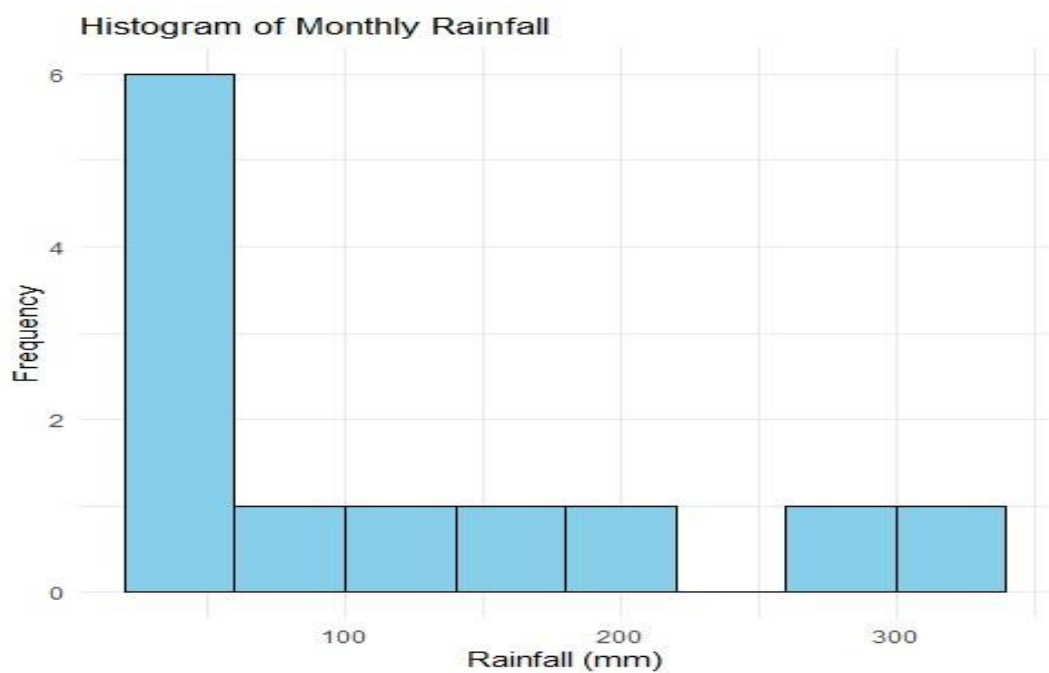
  Temperature = c(22,24,27,30,33,31,29,28,28,27,25,23),

  Humidity = c(60,62,65,68,75,85,90,88,84,78,70,65)

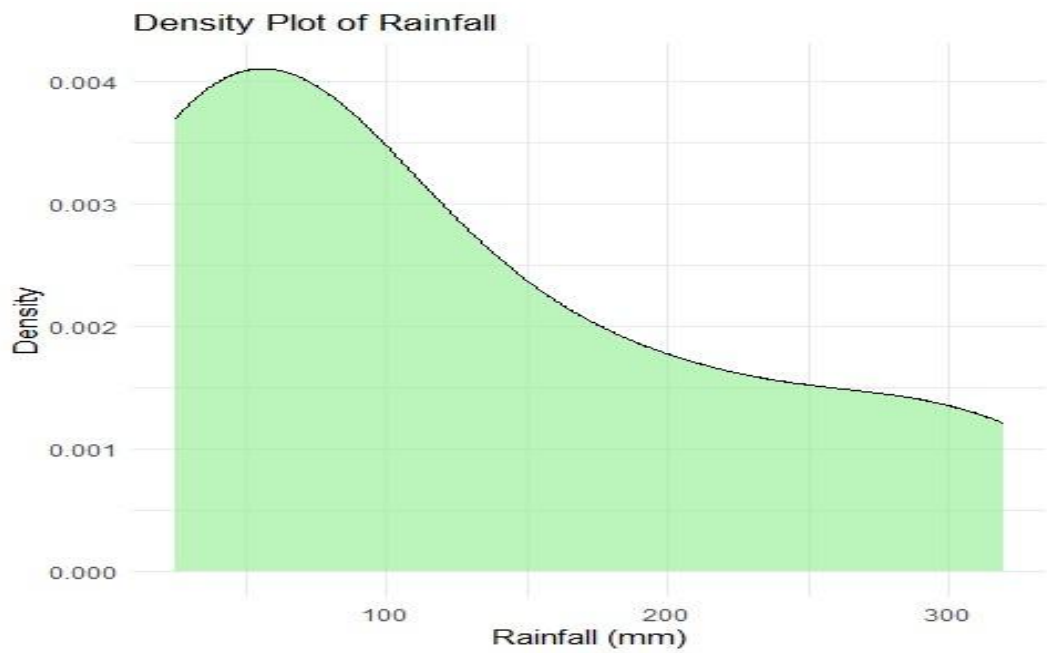
)
```

```
ggplot(climate,
aes(x=Rainfall))+
geom_histogram(binwidth=40,
               fill="skyblue",
               color="black")+ labs(title="Histogram of
Monthly Rainfall",    x="Rainfall (mm)",
y="Frequency")+ theme_minimal()
```

Output:



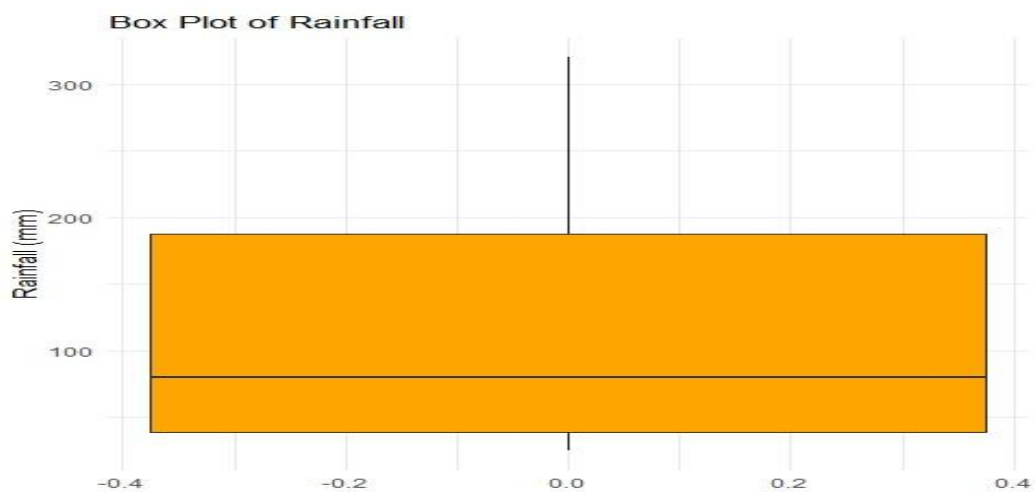
Density Plot: ggplot(climate,
aes(x=Rainfall))+
geom_density(fill="lightgreen",
 alpha=0.6)+
labs(title="Density Plot of Rainfall",
x="Rainfall (mm)", y="Density")+
theme_minimal() **Output:**



Box Plot:

```
ggplot(climate,
aes(y=Rainfall))+
geom_boxplot(fill="orange")+
labs(title="Box Plot of Rainfall",
y="Rainfall (mm)")+
theme_minimal()
```

Output:

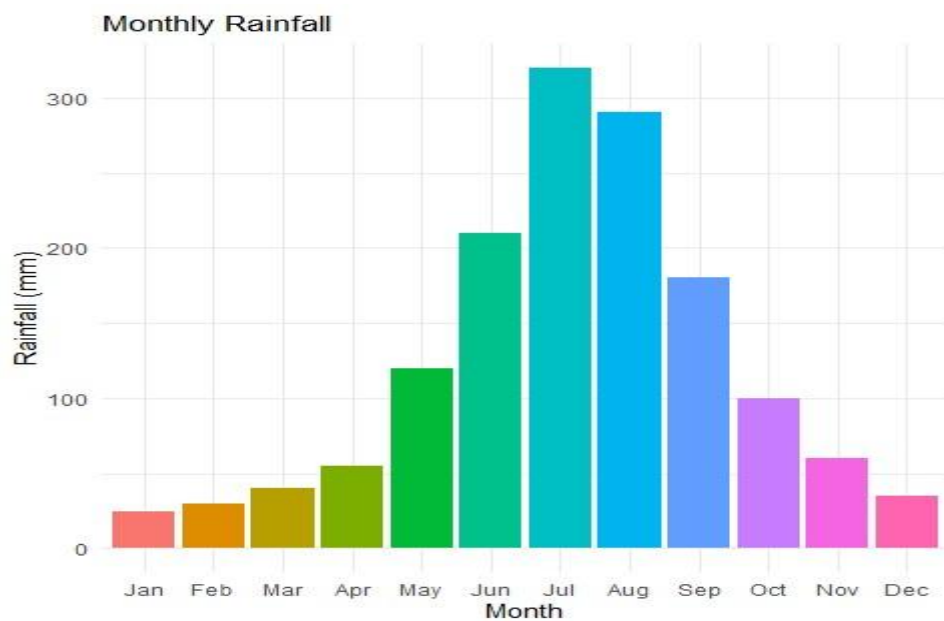


Bar Chart: ggplot(climate,

```
aes(x=Month,
y=Rainfall,
```

```
fill=Month)))+
geom_bar(stat="identity")+
labs(title="Monthly Rainfall",
x="Month", y="Rainfall
(mm)")+ theme_minimal()+
theme(legend.position="none")
```

Output:



4. Bank Loan Risk Assessment

Histogram: # Load Library

```
library(ggplot2) # Create Dataset
```

```
loan <- data.frame(
```

```
  Customer = paste0("C", 1:10),
```

```
  LoanAmount = c(150000, 200000, 120000, 300000, 250000,
```

```
    180000, 220000, 275000, 160000, 210000),
```

```
  Income = c(50000, 65000, 45000, 90000, 75000,
```

```
    60000, 70000, 85000, 55000, 68000),
```

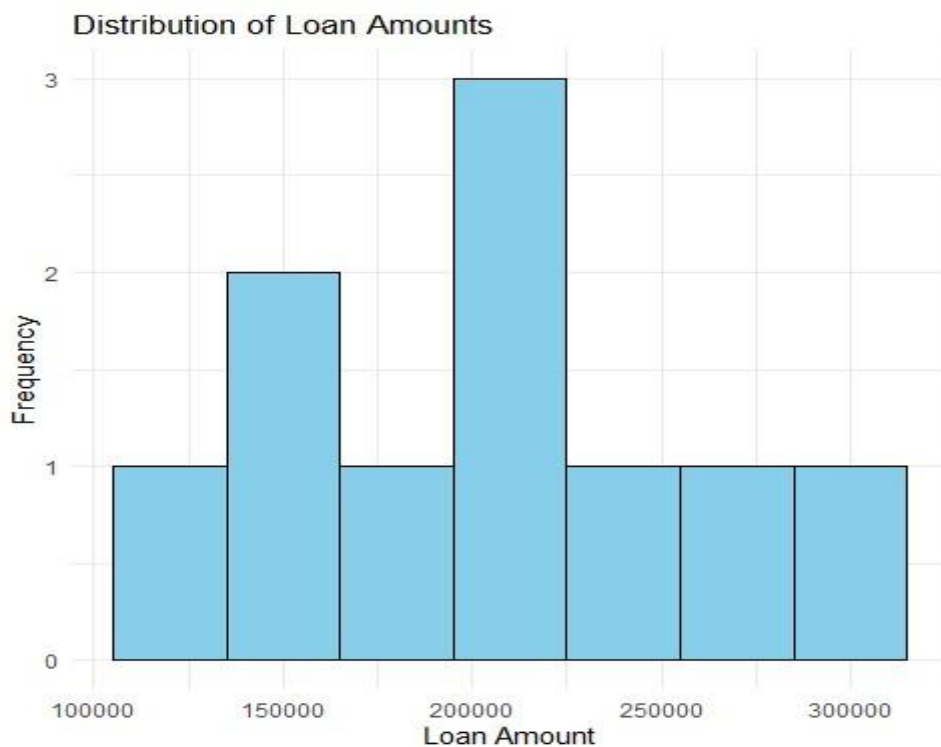
```
  CreditScore = c(720, 680, 750, 620, 700,
```

```
    690, 710, 640, 730, 695),
```

```

Repayment = c("Good","Average","Good","Poor","Good",
              "Average","Good","Poor","Good","Average")
)
ggplot(loan, aes(x =
LoanAmount)) +
geom_histogram(binwidth = 30000,
              fill = "skyblue", color =
"black") + labs(title = "Distribution of Loan
Amounts", x = "Loan Amount", y =
"Frequency") + theme_minimal() Output:

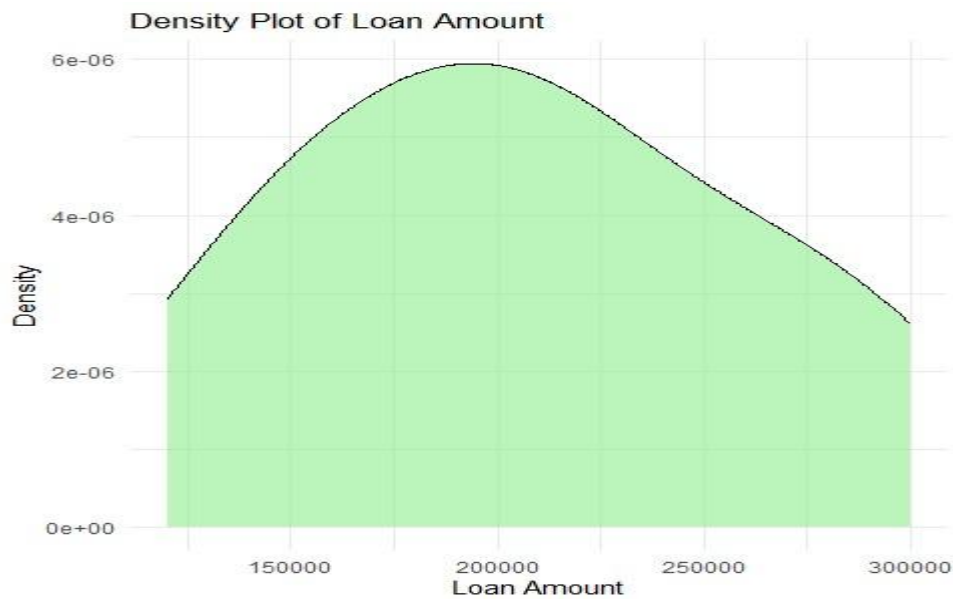
```



```

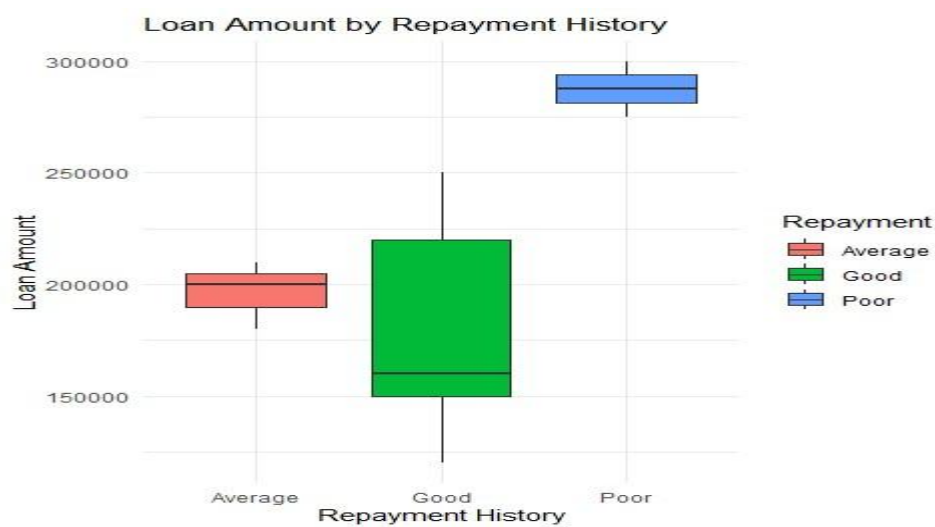
Density Plot: ggplot(loan,
aes(x = LoanAmount)) +
geom_density(fill = "lightgreen",
              alpha = 0.6) + labs(title =
"Density Plot of Loan Amount", x =
"Loan Amount", y = "Density") +
theme_minimal() Output:

```



Box Plot: `ggplot(loan, aes(x = Repayment, y = LoanAmount, fill = Repayment)) +
geom_boxplot() + labs(title = "Loan Amount by
Repayment History", x = "Repayment
History", y = "Loan Amount") +
theme_minimal()`

Output:



Scatter Plot: `ggplot(loan, aes(x = CreditScore, y = LoanAmount, color = Repayment)) + geom_point(size =`

```
4) + labs(title = "Loan Amount vs Credit
Score", x = "Credit Score", y =
"Loan Amount") + theme_minimal()
```

Output:



```
Bar Chart: ggplot(loan, aes(x =
Customer, y = LoanAmount,
fill = Repayment)) + geom_bar(stat =
"identity") + labs(title = "Customer
Loan Amount", x = "Customer",
y = "Loan Amount") +
theme_minimal()
```

Output:

